

MAHAJANGA, Madagascar

WMO: 670270

Lat: 15.667S

Lon: 46.352E

Elev: 27

StdP: 101.01

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	18.9	9.1	7.2	29.4	10.7	8.0	28.1	9.5	27.8	8.6	28.0	1.6	160	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	8.4	34.4	23.7	33.9	23.8	33.2	24.0	27.9	31.2	27.5	30.9	27.1	30.5	4.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	22.8	30.4	26.4	21.9	29.8	26.1	21.6	29.6	89.6	31.2	87.4	31.3	86.0	31.0	31.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.6	16.0	36.3	0.7	0.8	15.5	36.9	15.0	37.3	14.6	37.7	14.1	38.3		
(5)			13.4	29.1	0.7	0.6	12.9	29.6	12.5	30.0	12.1	30.3	11.5	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.2	27.7	27.9	28.4	28.4	27.1	25.6	25.0	25.6	26.5	27.8	28.5	28.2	(6)
(7)		DBStd	1.60	1.12	1.01	0.94	0.91	1.18	1.18	1.04	1.02	1.02	1.13	1.04	1.12	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6288	550	501	570	552	531	469	465	483	496	552	556	565	(10)
(11)		CDD18.3	3247	292	267	312	302	273	219	207	224	246	293	306	307	(11)
(12)		CDH23.3	34656	3177	2951	3588	3474	2862	2046	1816	2041	2333	3178	3592	3597	(12)
(13)		CDH26.7	13871	1134	1071	1466	1522	1210	804	673	804	942	1322	1504	1419	(13)
(14)	Wind	WSAvg	3.2	3.2	3.0	2.8	2.5	2.6	2.9	3.3	3.6	3.7	3.6	3.6	3.0	(14)
(15)	Precipitation	PrecAvg	1326	453	312	175	64	6	1	2	1	16	100	249		(15)
(16)		PrecMax	1761	863	1020	533	250	79	8	10	8	21	67	417	731	(16)
(17)		PrecMin	641	80	83	26	0	0	0	0	0	0	0	0	1	(17)
(18)		PrecStd	340	204	211	111	65	15	2	3	2	4	19	102	156	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	33.8	34.2	35.0	34.2	33.1	32.2	33.2	34.2	35.2	34.2	(19)	
MCWB			25.3	25.9	25.5	25.1	23.4	22.0	20.7	20.5	21.0	22.1	23.1	25.0	(20)	
2%		DB	32.8	32.6	33.5	34.0	33.2	32.2	31.8	32.2	33.1	34.0	34.0	33.1	(21)	
		MCWB	25.9	26.2	25.6	25.0	23.2	21.4	20.5	20.2	21.0	22.0	23.2	25.4	(22)	
5%		DB	31.9	32.0	32.9	33.2	32.8	31.6	31.0	31.8	32.2	33.0	33.0	32.2	(23)	
		MCWB	26.3	26.4	25.8	25.0	23.2	21.2	20.2	20.1	20.9	22.4	24.1	25.8	(24)	
10%		DB	31.1	31.1	32.0	32.7	32.0	30.9	30.2	30.9	31.2	32.0	32.1	31.8	(25)	
		MCWB	26.5	26.5	26.1	25.0	23.2	21.1	20.0	20.0	21.2	22.9	24.6	25.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.2	28.2	28.2	28.1	26.7	25.1	24.1	24.5	25.4	26.7	27.7	28.1	(27)
MCD B			31.2	31.0	31.6	31.9	30.8	28.7	27.7	28.1	29.3	30.9	31.2	31.6	(28)	
2%		WB	27.6	27.6	27.7	27.2	26.1	24.2	23.2	23.6	24.5	26.0	26.8	27.4	(29)	
		MCD B	30.7	30.7	31.1	30.9	29.8	27.8	26.9	27.3	28.6	29.9	30.6	31.0	(30)	
5%		WB	27.1	27.2	27.3	26.7	25.5	23.6	22.6	23.0	23.9	25.5	26.4	26.9	(31)	
		MCD B	30.3	30.3	30.6	30.3	29.3	27.4	26.7	26.8	27.9	29.4	30.1	30.4	(32)	
10%		WB	26.7	26.7	26.8	26.2	24.9	22.9	22.0	22.4	23.4	25.0	26.0	26.5	(33)	
		MCD B	29.8	29.8	30.1	29.8	28.7	27.1	26.5	26.7	27.5	28.9	29.7	29.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.6	6.7	7.8	9.3	10.5	11.4	11.7	11.9	11.4	10.1	8.4	7.1	(35)
MCDBR			7.8	7.8	8.9	9.9	11.0	11.7	12.1	12.6	12.2	11.3	9.6	8.1	(36)	
MCWBR			3.9	3.8	4.1	4.4	4.7	5.0	5.2	5.7	5.8	5.5	4.7	3.9	(37)	
5% WB		MCDBR	6.9	6.9	7.6	8.8	10.0	10.9	11.4	11.6	11.0	9.3	7.7	7.1	(38)	
		MCWBR	3.9	3.9	4.1	4.3	4.9	5.3	5.6	5.9	5.7	4.8	4.0	3.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.424	0.414	0.387	0.358	0.343	0.326	0.327	0.340	0.376	0.417	0.452	0.427	(40)	
taud		2.476	2.503	2.585	2.662	2.665	2.691	2.667	2.593	2.462	2.379	2.322	2.444	(41)		
Ebn at Noon		922	923	928	922	903	904	911	926	924	909	890	919	(42)		
Edh at Noon		118	114	102	89	83	78	82	94	114	128	137	122	(43)		
(44)	All-Sky Solar Radiation	RadAvg	5.55	5.76	6.04	5.88	5.27	4.93	5.10	5.83	6.67	7.11	6.83	6.20	(44)	
(45)		RadStd	0.59	0.56	0.29	0.20	0.16	0.07	0.07	0.07	0.10	0.17	0.28	0.42	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.24	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page